QUICK REFERENCE

KEY	IN THE TERMINAL
`BREAK`	Menu
`CAPS`+`1`	Esc
`CAPS`+`2`	Caps lock
`CAPS`+`3`	Tab
`CAPS`+`5`–`8`	Arrows
`CAPS`+`9`	32 / 64 / 80
`CAPS`+`0`	Backspace
`CAPS`+`SYM`, key	Ctrl-key
`SYM`+`A` `S` `D`	`~ | \`
`SYM`+`F` `G` `H`	`{ } ^`
`SYM`+`Y` `U` `X`	`` [ ] ` ``

KEY	IN THE MENU
`D`	Phonebook
`H`	Hang up
`W`	WiFi
`S`	Settings (`S` saves)
`X`	XMODEM download
`C`	Clear the screen
`Q`	Quit to BASIC
`BREAK`	Back

DIALLING BY HAND

`ATDT"host:port"` telnet · `ATD"host:port"` raw · `ATI` status · `ATW` scan WiFi

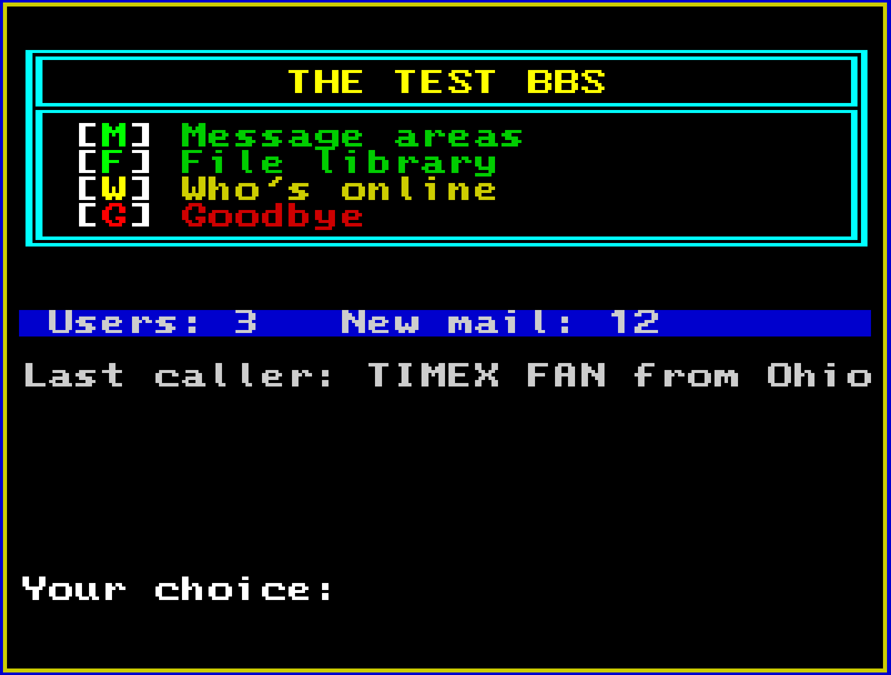
PicoTerm 0.6 © 2026 · for the TS-2068 and TS-Pico v3 · timexsinclair.com

TIMEX/SINCLAIR 2068 · TS-PICO V3 · 2026

PICOTERM

USER MANUAL

A terminal program for the TS-2068 and the TS-Pico v3 card. The card's WiFi modem runs Zimodem, and PicoTerm uses it to reach telnet BBSes: ANSI colour and box drawing, 32 or 64 columns (80 with PicoVideo), a phonebook, and XMODEM downloads to the SD card.

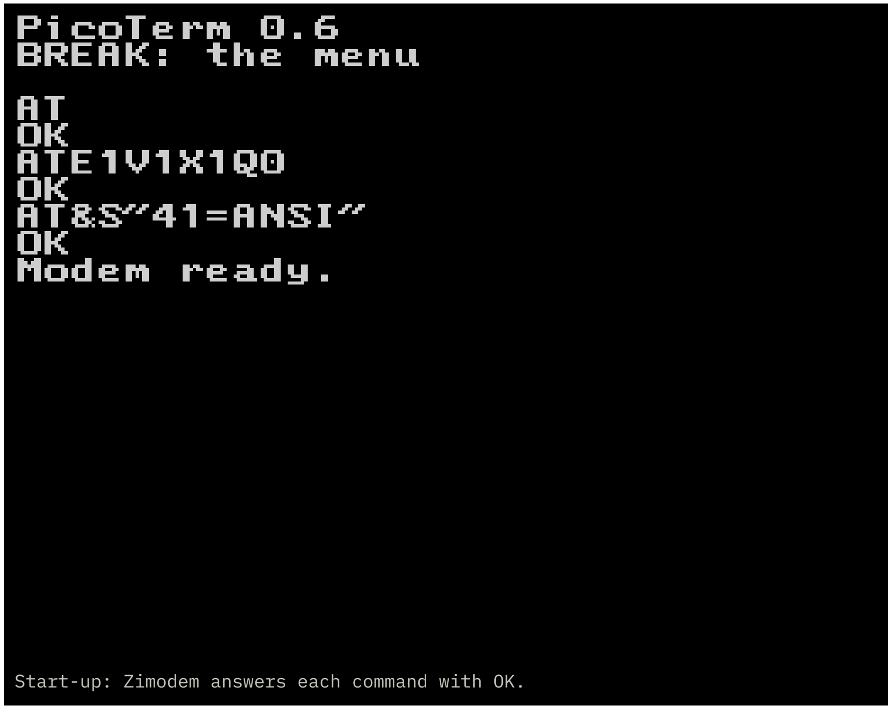


```
THE TEST BBS
[M] Message areas
[F] File library
[W] Who's online
[G] Goodbye

Users: 3    New mail: 12
Last caller: TIMEX FAN from Ohio

Your choice:
```

TS-2068 + TS-Pico v3 · WiFi

Version 0.6, 2026

timexsinclair.com



```
PicoTerm 0.6
BREAK: the menu

AT
OK
ATE1V1X1Q0
OK
AT&S"41=ANSI"
OK
Modem ready.
```

Start-up: Zimodem answers each command with OK.

PicoTerm 0.6 © 2026. Written in Z80 assembly for the TS-Pico v3 card.

Built on Bo Zimmerman's Zimodem, the TS-Pico firmware and ROM, and the TS 2068 Technical Manual's Video Mode Change Service (as Zebra's OS-64 uses it). The CP437 font is PicoVideo's: Bm437_ACM_VGA_8x8 from VileR's Ultimate Oldschool PC Font Pack (int10h.org), CC BY-SA 4.0.

This manual was written from the program itself. Its screenshots come from PicoTerm's own code run on a model of the 2068 with a simulated modem and BBS, so the BBS screens are examples.

`picoterm.tap: loader CLEAR 34815: LOAD "" CODE: RANDOMIZE USR 34816 · code at 8800h · 8251 $73/$77 · TS-Pico 14/15 · AY F5h/F6h`

YOUR BBS LIST

NAME	HOST:PORT	TELNET	COLS	NOTES

13 LIMITS

- **Downloads only, and at most 6,912 bytes:** no upload, YMODEM or ZMODEM, and no capture to a file.
- **No scrollback.** Text that scrolls off the top is gone.
- **24 rows.** BBSes that assume 25 lose the bottom line.
- **Colour:** no bright background on its own; 64 columns on a stock 2068 has one colour pair.
- **Keyboard:** the symbol keys follow the Spectrum's legends, which the 2068 mostly shares; the tables in section 4 are what PicoTerm does.
- **Screen width:** PicoTerm can't tell a telnet host your width itself. Most BBSes ask, or measure it with the cursor-position question, which PicoTerm answers.
- **The modem:** the TS-Pico's, in Hayes mode, with its 8251 emulation. Not a stock TS2050.

14 ABOUT THIS EDITION

PicoTerm 0.6, 2026, for the TS-Pico v3 card, in Z80 assembly (sjasmplus). This manual was written from the program itself; the BBS screens are examples from a simulated BBS, not real systems. Credits are inside the front cover.

CONTENTS

01 What you need 4
02 Loading PicoTerm 4
03 Your first call 6
04 The keyboard 7
05 The terminal 9
06 32, 64 and 80 columns 10
07 The menu 11
08 The phonebook 12
09 WiFi 14
10 Settings 15
11 Downloading files 16
12 Messages and problems 17
13 Limits 18
14 About this edition 18
Your BBS list 19

KEYS IN THIS MANUAL

`CAPS` is CAPS SHIFT and `SYM` is SYMBOL SHIFT. `BREAK` is `CAPS`+`SPACE`; it opens PicoTerm's menu, and in any PicoTerm screen it goes back one step.

01 WHAT YOU NEED

- **A TS-2068** with the TS-Pico ROM 2.3 or later.
- **The TS-Pico v3 card**, with firmware that has the TS2050 modem emulation (the 8251 at ports `$73` and `$77`), and Zimodem on the card's ESP32. The card's modem has to be in *Hayes mode*. PicoTerm switches it there itself where the firmware allows.
- **A WiFi network** the card can join. PicoTerm can scan for one and join it (section 9).
- **An SD card in the TS-Pico**, for `picoterm.cfg` (your settings and phonebook) and for downloads.
- *Optional:* **PicoVideo**, for 80 columns and for colour in 64 columns, on its HDMI output.

A REAL TS2050?

PicoTerm is written for the TS-Pico's modem. It doesn't pulse-dial, and it doesn't watch the TS2050's carrier line, so it won't drive an original 300-baud TS2050. Use MTERM for that.

02 LOADING PICOTERM

`picoterm.tap` holds a one-line BASIC loader and about 15 KB of machine code, which loads at 34816 (8800h).

1. Copy `picoterm.tap` to the SD card's `/TAP` folder.
2. Type `LOAD "tpi:picoterm.tap"` `ENTER` to mount it.
3. Type `LOAD ""` `ENTER`. The loader runs by itself.

12 MESSAGES AND PROBLEMS

YOU SEE	IT MEANS, AND WHAT TO DO
`Modem ready.`	Zimodem answered. Type or dial.
`This TS-Pico has no tpi:modem.`	Older card firmware. Set the card's modem to Hayes mode yourself (see the TS-Pico manual), or update.
`No modem: the 8251 isn't ready.`	Nothing at `$73` / `$77` within 5 s: no v3 card, the modem off, or no modem firmware. PicoTerm runs; modem keys are dropped.
`No answer from the modem.`	No `OK` in about 12 s. Check the modem is in **Hayes** mode (TS2050 mode hides Zimodem's replies), then type `AT` `ENTER`.
`NO ANSWER`, `ERROR`	Zimodem couldn't connect, or didn't understand: check the host, port, quotes and WiFi (`ATI` shows Zimodem's state).
`NO CARRIER`	The host hung up.
`Hang up first.`	Dialling or WiFi while online: `BREAK`, `H`.
`Couldn't save the settings.`	No TS-Pico or SD card, or the card refused.
`Too big for memory (6912).`	The download was too long for this version.
`The sender didn't start.`	Nothing from the BBS for about 30 s. Start the download on the BBS first.
`Too many errors. Cancelled.`	Ten bad blocks in a row. Try again.

SLOW TEXT

The card passes text to the 2068 no faster than its `MODEM_BAUD` setting: 300 baud unless changed, which is 30 characters a second. Raise it in the TS-Pico's `config.ini` (2400 or more); PicoTerm keeps up.

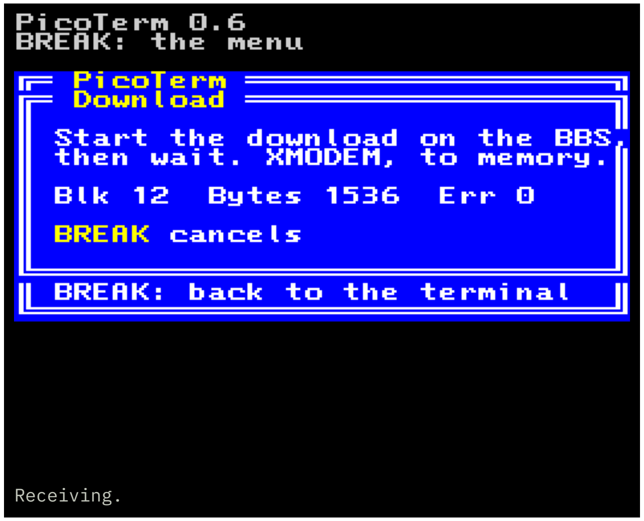
11

DOWNLOADING FILES

PicoTerm receives with **XMODEM**, CRC or 1K, with checksums for older senders. The file goes to the TS-Pico's SD card.

1. On the BBS, choose the file and XMODEM.
2. `BREAK`, `X`, and a name for the file (`download.bin`). PicoTerm asks before replacing a file.
3. The window counts blocks, bytes and errors. Damaged blocks are asked for again.
4. **Strip ^Z padding?** XMODEM pads the last block with 1Ah. `Y` for text; `N` for programs and binaries.
5. PicoTerm saves the file and shows its size.

`BREAK` cancels and tells the BBS. In stock 64-column mode, PicoTerm uses 32 columns during the transfer: the file is held in the 64-column screen's memory.



```
PicoTerm 0.6
BREAK: the menu

  PicoTerm
  Download

 Start the download on the BBS,
 then wait. XMODEM, to memory.

 Blk 12  Bytes 1536  Err 0

 BREAK cancels

 BREAK: back to the terminal





Receiving.
```

IN THIS VERSION

Downloads are received into memory first: at most **6,912 bytes**. The file is saved as `SAVE "f:name" CODE` saves it, so it starts with a 128-byte +3DOS header. `LOAD "f:name" CODE` loads it back; on a PC, cut the first 128 bytes. A planned TS-Pico firmware addition should lift both limits.

WHAT HAPPENS AT START-UP

1. PicoTerm reads `picoterm.cfg` from the card's current folder, if there is one, and sets up the screen in its column mode.
2. It puts the card's modem in Hayes mode (`tpi:modem hayes`).
3. It resets the 8251, which starts the card's ESP32.
4. It sends `AT` once a second until Zimodem answers `OK` (up to about 12 s; Zimodem's banner may show first), then your init string (`ATE1V1X1Q0`) and the telnet terminal type (`AT&S"41=ANSI"`).
5. It shows `Modem ready.` You see everything the modem says. If something is missing, a message says so and PicoTerm carries on (section 12).

HAYES MODE STAYS

The TS-Pico saves the modem's mode in `config.ini`, so the card stays in Hayes mode after you quit. Before loading MTERM or another TS2050 program, type `SAVE "tpi:modem 2050"`.

MEMORY

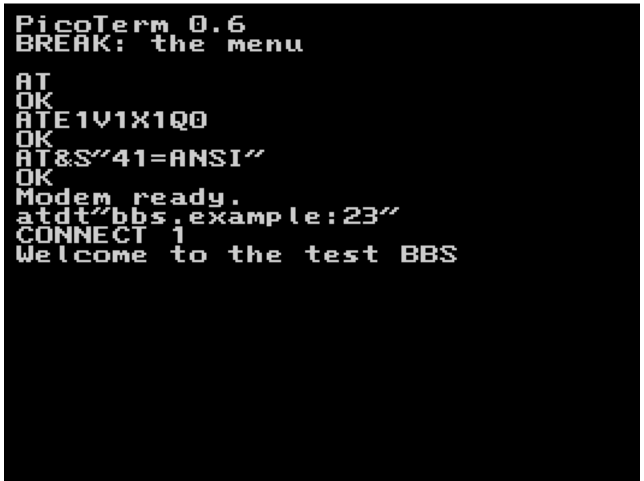
The loader does `CLEAR 34815`. PicoTerm also opens the second display file with the ROM's Video Mode Change Service, as OS-64 does, which moves BASIC up to 7B00h. It stays open after you quit, so BASIC has about 7 KB less room until you reset.

03 YOUR FIRST CALL

Once the modem is ready, everything you type goes to it. Zimodem echoes your typing and answers in words (`OK`, `CONNECT`, `NO CARRIER`), like a Hayes modem.

1. **Join your WiFi**, once: `BREAK`, then `W` (section 9).
2. **Dial a host** by typing its address in quotes: `ATDT"bbs.example.com:23"` `ENTER`. The `T` asks for telnet, which most BBSes want; `ATD"…"` is a raw TCP port.
3. Zimodem answers `CONNECT`. From now on everything you type goes to the BBS.
4. **To hang up**: `BREAK`, then `H`. If the BBS hangs up first, you see `NO CARRIER`.

Put the hosts you use in the phonebook (section 8) and dial them with one key.



```
PicoTerm 0.6
BREAK: the menu

AT
OK
ATE1V1X1Q0
OK
AT&S"41=ANSI"
OK
Modem ready.
atdt"bbs.example:23"
CONNECT 1
Welcome to the test BBS
```

Dialling by hand: ATDT, CONNECT, the BBS's greeting.

10 SETTINGS

`BREAK`, then `S`. Each key changes one setting at once.

KEY	SETTING	DEFAULT
`E`	Emulation	ANSI
`C`	Columns at start-up	32
`P`	PicoVideo fitted	No
`I`	64-column ink	White
`B`	Backspace: BS or DEL	BS
`L`	Local echo	No
`N`	ENTER sends CR or CR LF	CR
`F`	Blink shows as FLASH	Yes
`G`	Bell	Yes
`M`	Modem init string	`ATE1V1X1Q0`
`S`	**Save** settings and phonebook	

PICOTERM.CFG

`S` writes `picoterm.cfg` in the SD card's current folder; PicoTerm reads it at start-up. It's plain text: `emul=2` (0 TTY, 1 VT52, 2 ANSI), `cols=1` (0 is 32, 1 is 64, 2 is 80), `picovideo`, `ink64`, `bs` (8 or 127), `echo`, `enter`, `blink`, `bell`, `init`, and a line per phonebook entry:

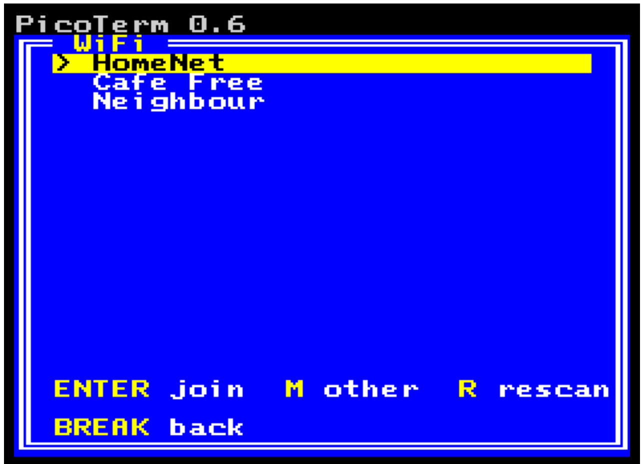
```
phone=Test BBS|bbs.example.com:23|1|2|0|8|0
```

That's the name, host, telnet, emulation, columns, backspace and echo. Lines PicoTerm doesn't understand are skipped.

09 WIFI

`BREAK`, then `W`, while offline. Zimodem scans (`ATW`) and PicoTerm lists up to 16 networks.

1. Choose one with the arrows and press `ENTER`, or `M` to type a name that isn't listed. `R` scans again.
2. Type the password (it shows as stars) and press `ENTER`. PicoTerm sends `ATW"name,password"` and waits up to 20 s.
3. If it worked, `Y` makes it the default (`AT&W`): Zimodem joins it by itself from then on.



```
PicoTerm 0.6
  WiFi
 > HomeNet
   Cafe Free
   Neighbour

 ENTER join   M other   R rescan
 BREAK back
```

The networks Zimodem can see.

COMMAS AND QUOTES

Zimodem's `ATW"name,password"` can't carry a comma or a `"` in either part.

04 THE KEYBOARD

Letters, digits and `SPACE` type as you'd expect; `CAPS` gives capitals. Every other ASCII character and control key is on a shifted key. A held key repeats after about half a second.

KEYS	LEGEND	SENDS OR DOES
`CAPS`+`1`	EDIT	**Esc** (1Bh)
`CAPS`+`2`	CAPS LOCK	Caps lock, for letters
`CAPS`+`3`	TRUE VID	**Tab** (09h)
`CAPS`+`5`–`8`	← ↓ ↑ →	**Arrows:** `ESC [ D B A C` (ANSI, TTY), `ESC D B A C` (VT52). In PicoTerm's screens they move the selection.
`CAPS`+`9`	GRAPHICS	**Next column mode:** 32, 64, 80
`CAPS`+`0`	DELETE	**Backspace:** BS (08h) or DEL (7Fh), as set
`BREAK`	BREAK	**The PicoTerm menu;** in a menu, back
`ENTER`		CR, or CR LF as set

CONTROL KEYS

Press `CAPS` and `SYM` together, let go, then press a letter: PicoTerm sends it as a control key. `CAPS`+`SYM`, `C` sends **Ctrl-C.** Pressing `CAPS`+`SYM` twice cancels it. It works for the letters and `[` `\` `]` `^` `_`.

SYMBOL SHIFT

The red symbols work as in BASIC. The keys that type keywords give the ASCII characters the 2068 can't otherwise type:

KEY	SENDS	KEY	SENDS	KEY	SENDS
`SYM`+`A`	`~`	`SYM`+`S`	`|`	`SYM`+`D`	`\`
`SYM`+`F`	`{`	`SYM`+`G`	`}`	`SYM`+`H`	`^`
`SYM`+`Y`	`[`	`SYM`+`U`	`]`	`SYM`+`X`	`` ` `` (not £)

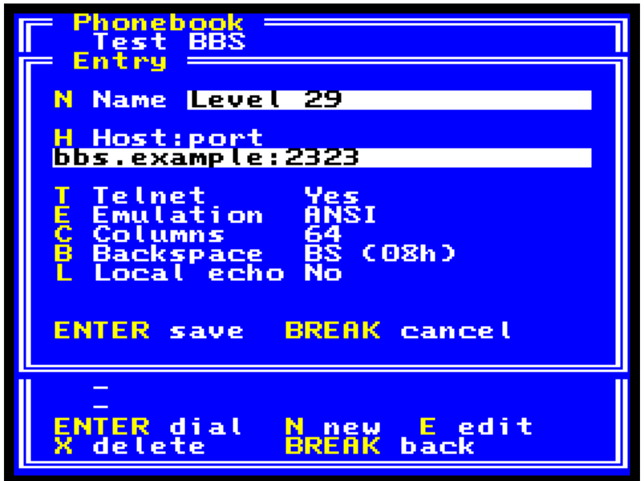
KEY	SENDS	KEY	SENDS	KEY	SENDS
`SYM`+`Z`	`:`	`SYM`+`C`	`?`	`SYM`+`V`	`/`
`SYM`+`R`	`<`	`SYM`+`T`	`>`	`SYM`+`O`	`;`
`SYM`+`P`	`"`	`SYM`+`J`	`-`	`SYM`+`K`	`+`
`SYM`+`L`	`=`	`SYM`+`B`	`*`	`SYM`+`N`	`,`
`SYM`+`M`	`.`	`SYM`+`1`…`0`:	`! @ # $ % & ' ( ) _`		

`SYM` with `Q`, `W`, `E` or `I` (≤ ≠ ≥ AT) sends nothing.

AN ENTRY

KEY	FIELD	NOTES
`N`	Name	Up to 16 characters
`H`	Host:port	Up to 40, e.g. `bbs.example.com:23`
`T`	Telnet	Yes: `ATDT`. No: `ATD`, raw TCP
`E`	Emulation	ANSI, VT52, TTY
`C`	Columns	32, 64, 80 (64 without PicoVideo)
`B`	Backspace	BS or DEL, as the host wants
`L`	Local echo	For hosts that don't echo

`ENTER` saves the entry; `BREAK` drops the changes. In a text field `DELETE` rubs out and `ENTER` accepts.



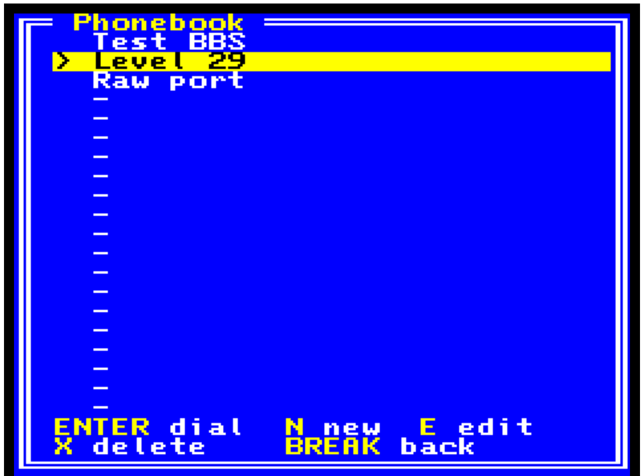
```
  Phonebook
   Test  BBS
  Entry

 N  Name  Level 29

 H  Host:port
bbs.example:2323

 T  Telnet       Yes
 E  Emulation    ANSI
 C  Columns      64
 B  Backspace    BS (08h)
 L  Local echo   No


 ENTER save   BREAK cancel

    _
    _
 ENTER dial   N new  E edit
 X delete     BREAK back
```

Editing an entry.

08 THE PHONEBOOK

20 hosts. Move with `CAPS`+`6` and `CAPS`+`7`.

KEY	DOES
`ENTER`	**Dial**: applies the entry's settings, goes back to the terminal, sends `ATDT"host:port"` (`ATD` with telnet off).
`N`	**New**: the name, `ENTER`, the host, `ENTER`, then `ENTER` to save. It starts with your current settings.
`E` / `X`	**Edit** / **delete** (`Y` confirms).



```
 Phonebook
  Test BBS
> Level 29
  Raw port
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
  -
ENTER dial   N new   E edit
X delete     BREAK back
```

The phonebook.

REMEMBER TO SAVE

Entries live in memory until you save your settings (Settings `S`). Dialling while online is refused: “Hang up first.”

05 THE TERMINAL

PicoTerm has three emulations. It uses ANSI unless you choose otherwise in Settings or in a phonebook entry, and it tells Zimodem which one to report to telnet hosts.

EMULATION	WHAT IT UNDERSTANDS
ANSI	For BBSes: a VT100 subset with colour, as BBS software expects of "ANSI-BBS". • Cursor moves, positioning, save and restore. • Erasing; inserting and deleting lines and characters. • A scrolling region. • Colours with bold, blink and reverse. • It answers the cursor-position question, which BBSes use to find the screen size, and identifies itself as a VT100.
VT52	The standard set: cursor moves, home, `ESC Y` addressing, erasing, reverse line feed, identify.
TTY	Text and control codes only; escape sequences show as text.

In every emulation PicoTerm handles **BEL**, a short tone on the AY; **BS**; **Tab**, every 8 columns; **LF**; **CR**; and **FF**, which clears the screen. Text wraps as on a VT100: a character in the last column leaves the cursor there, and the next starts a new line.

COLOUR AND CHARACTERS

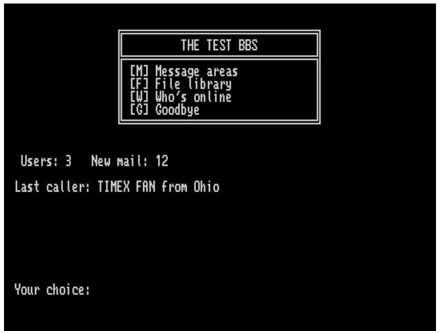
ANSI's eight colours become the 2068's. **Bold** is BRIGHT, **blink** is FLASH (or not, in Settings), and **reverse** swaps ink and paper. BRIGHT lights the whole cell, so bold text on a coloured background brightens the background too.

PicoTerm uses the PC's CP437 characters, as PicoVideo does for 80 columns, so box lines, shading and accented letters look as BBS authors meant.

06 32, 64 AND 80 COLUMNS

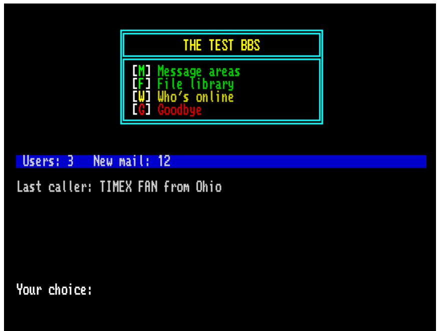
`CAPS`+`9` changes mode, keeping what's on screen. Settings `C` picks the mode PicoTerm starts in.

MODE	NEEDS	COLOUR
32	any 2068	Per character. Readable on any TV.
64	any 2068	One ink and paper (Settings `I`); per character with PicoVideo.
80	PicoVideo	Per character, on HDMI. The 2068's own video shows nothing useful. Turn on **PicoVideo** in Settings.



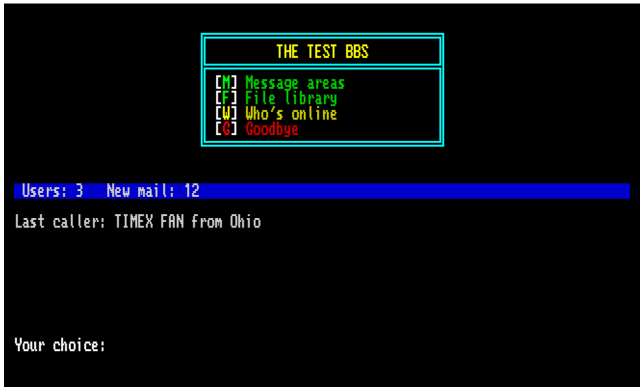
```
THE TEST BBS
[M] Message areas
[F] File library
[W] Who's online
[G] Goodbye

 Users: 3   New mail: 12
Last caller: TIMEX FAN from Ohio

Your choice:
```

64 columns, stock 2068.



```
THE TEST BBS
[M] Message areas
[F] File library
[W] Who's online
[G] Goodbye

 Users: 3   New mail: 12
Last caller: TIMEX FAN from Ohio

Your choice:
```

64 with PicoVideo.



```
THE TEST BBS
[M] Message areas
[F] File library
[W] Who's online
[G] Goodbye

 Users: 3   New mail: 12
Last caller: TIMEX FAN from Ohio

Your choice:
```

80 columns, PicoVideo's text mode.

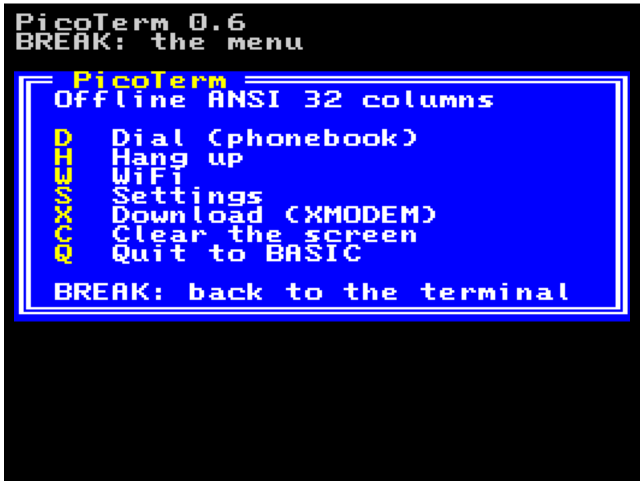
The screen is 24 rows. A line scrolls in about 40 ms in 32 and 80 columns and 100 ms in 64: fast enough for 2400 baud.

07 THE MENU

Press `BREAK` at any time. The menu's top line says whether you're online, and the emulation and columns in use.

KEY	DOES
`D`	The phonebook: dial, add, edit, delete
`H`	Hang up
`W`	WiFi: scan, join, save
`S`	Settings
`X`	Download a file (XMODEM)
`C`	Clear the screen
`Q`	Quit to BASIC; hangs up first if online
`BREAK`	Back to the terminal

Text from the BBS waits while a menu is open, and appears when you return.



```
PicoTerm 0.6
BREAK: the menu

 PicoTerm
 Offline ANSI 32 columns

 D  Dial (phonebook)
 H  Hang up
 W  WiFi
 S  Settings
 X  Download (XMODEM)
 C  Clear the screen
 Q  Quit to BASIC

 BREAK: back to the terminal
```

The menu.